SUNFLASER
OPTOELECTRONIC SOLUTIONS

SFA1000D

Laser Rangefinder Module

Product Manual

Version 202506V1

Disclaimer

This product manual is intended to provide reference specifications and performance parameters of the product. If certain specifications are not included in this document, please contact SunFlaser Technology (Chengdu) Co., Ltd. directly for further confirmation. All contents of this manual and the specifications of the module are subject to change without prior notice.

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Safety Instructions

For the safe use of the module, please carefully follow the safety instructions provided in this manual. Do not attempt to disassemble, modify, or repair any part of the module. Do not tamper with or attempt to alter the performance of the module.

Intended Use

- Take precautions against electrostatic discharge (ESD). Do not touch any electronic components or parts of the module without proper protective measures.
- Keep this manual readily accessible to all personnel operating the module.
- Operate the module only within the specified voltage and power range.
- Operate the module in a clean, dry, and ESD-protected environment.
- Do not touch the optical window lens with your fingers.
- Avoid exposing the module to environments with drastic temperature fluctuations.

Improper Use

- Using the module with the host system before verifying that all functions are operating properly;
- Operating the module without first reading and understanding this manual and all safety instructions;
- Disassembling, modifying, or altering the module without authorization from SunFlaser Technology (Chengdu) Co., Ltd.;
- Sending operating commands other than those specified in this manual;
- Using or testing the module in non-standard, abnormal, or unsuitable environments;
- Operating the module in a manner inconsistent with the instructions provided in this manual;

- Continuing to operate the module after improper handling or misuse;
- Using a module with obvious faults, defects, or damage;
- Performing test operations within the minimum ranging distance of the module.

Improper use may result in the following risks:

- Eye injury
- Incorrect results
- Measurement errors
- Property damage
- Equipment malfunction

Operator Responsibilities

- Be familiar with the basic knowledge required for the correct operation of the module;
- Understand and comply with the operation and maintenance requirements of the module;
- Prevent damage to the module during storage and transportation;
- Keep the module clean and well maintained;
- Ensure that the module is stored and operated in a safe environment;
- Do not expose the module to strong mechanical shock or impact;
- Do not expose the module to environments with drastic temperature fluctuations.

Avoid Measurement Errors

- Pay attention to all factors that may affect the performance of the module;
- After the Laser Rangefinder Module has been subjected to improper handling (such as vibration, impact, or dropping), or before performing critical measurement tasks, it is recommended to verify the performance specifications of the module.

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01 Product Overview

The SFA1000D Miniature Laser Rangefinder Module (hereinafter referred to as the "module") is a compact laser ranging device designed for precise distance measurement in lightweight and space-constrained applications.

The module operates at a wavelength of 905 nm and complies with IEC 60825-1 Class 1 laser safety standards under normal operating conditions. It features compact size, low power consumption, stable performance, and easy system integration through a UART-TTL communication interface.

With a modular optical and electronic design, the SFA1000D is suitable for applications including UAV payloads, handheld devices, outdoor observation equipment, and compact electro-optical systems.

02 Key Features

• Long-Range Measurement:

Maximum ranging distance up to 1000 m under standard test conditions

• High Accuracy:

Typical ranging accuracy of ±1 m within 400 m

• Dual Measurement Modes:

Supports both single-shot and continuous ranging modes

• UART-TTL Communication:

Simple serial interface for rapid system integration

• Optical Axis Calibration Mode:

Convenient for assembly alignment and debugging

• Compact and Lightweight:

Diameter 17 mm, length 37.2 mm, weight ≤10 g

• Low Power Consumption:

Supports 3.3 V to 5 V DC power input

03

Key Specifications

No.	Parameter	Specification
1	Laser Wavelength	905 nm
2	Laser Safety Classification	Class 1
3	Transmitting Aperture	Ø 4.2 mm
4	Receiving Aperture	Ø 14.5 mm
5	Maximum Measuring Range Test Conditions	≥1000 m Target: Large target; Visibility: ≥10 km; Target Reflectivity: 0.3
6	Minimum Measuring Distance (minimum ranging distance)	≤5 m
7	Measuring Accuracy	±1 m (≤400 m) ±0.4% of measured distance (>400 m)
8	Measurement Success Rate	Measurement Success Rate ≥99% (Test conditions: visibility ≥10 km, target reflectivity 30%)
9	Measuring Mode	Single Measurement / Continuous Measurement
10	Weight	≤10 g
11	Dimensions	Ø 17 mm × 37.2 mm
12	Electrical Interface	Connector Receptacle: 0.8WTB-6AB-01 Connector Plug: 0.8WTB-6Y-2
13	Supply Voltage	3.3 V – 5 V DC
14	Standby Power Consumption	≤0.6 W
15	Average Power Consumption	Average Power Consumption Short-range measurement: <0.85 W Long-range measurement: <1.35 W
16	Operating Temperature	-20°C to +60°C
17	Main Function	Laser Distance Measurement

Note:

The nominal ranging performance and accuracy are achieved under environmental conditions with visibility not less than 10 km and relative humidity not exceeding 60%. The ranging capability and accuracy of the Laser Rangefinder Module may vary depending on environmental conditions and target characteristics. Actual performance shall be subject to practical measurement results.

When the Laser Rangefinder Module is mounted on a stabilized platform, ensure that the platform can maintain stable target tracking. Otherwise, the ranging capability and measurement accuracy of the Laser Rangefinder Module may be affected.

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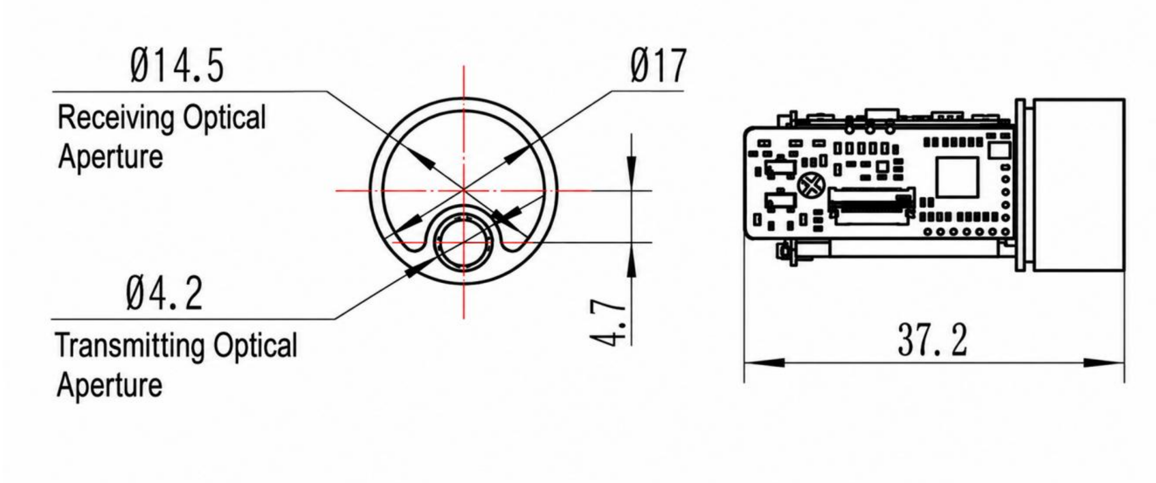
Interface and Connection

Mechanical and Optical Interfaces

The mechanical and optical interface dimensions of the Laser Rangefinder Module are shown in the figure below. Unit: mm.



Electrical Interface

The electrical interface requirements are as follows:

- Supply Voltage: 3.3 V – 5 V DC
- Communication Type: UART-TTL
- Standby Power Consumption: ≤0.6 W
- Average Power Consumption:

Short-range / minimum ranging distance: <0.85 W

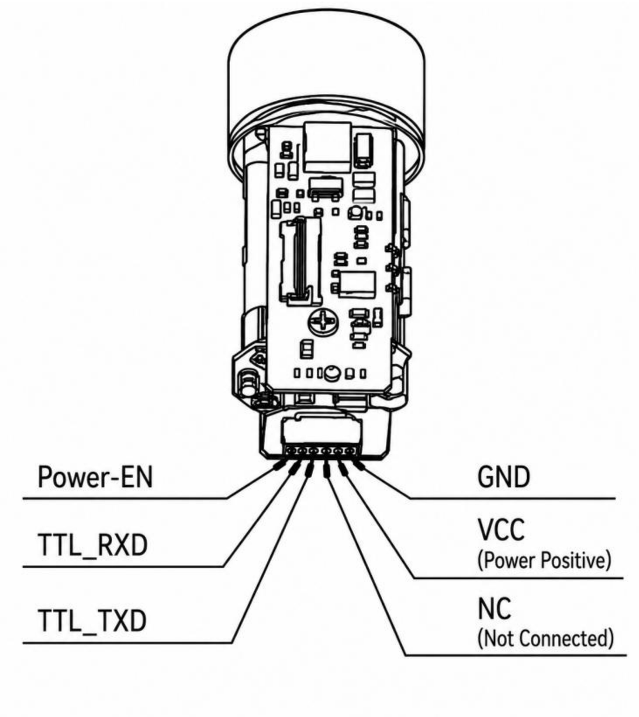
Long-range / Air target: <1.35 W

- Leakage Current: <100 μA

The host system is connected to the rangefinder through the 0.8WTB-6Y-2 connector, interfacing with the 0.8WTB-6AB-01 connector on the Laser Rangefinder Module side for interconnection testing.

The power supply and communication interface pin definitions of the Laser Rangefinder Module are shown in the table below.

Pin	Name	Type	Description
1	Power-EN	Input	Power enable control signal
2	TTL_RXD	Input	UART-TTL serial receive data
3	TTL_TXD	Output	UART-TTL serial transmit data
4	NC	—	Not connected
5	VCC	Power	3.3–5 V DC power input
6	GND	Ground	System ground



Note:

Power Management Logic

After VCC power is applied, the module enters standby mode.

When the Power-EN pin is pulled high, the module enters active mode and enables laser operation.

When the Power-EN pin is pulled low, the module enters low-power mode with leakage current less than 100 μA.

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The optical axis calibration mode can be activated through the corresponding communication command.

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Communication Protocol and Commands

For detailed information, please refer to Appendix 1 for the following contents:

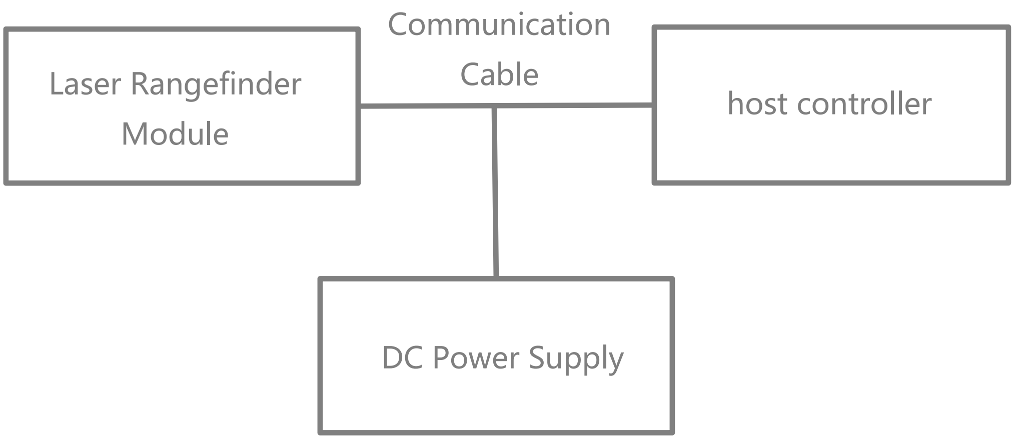
- Communication Interface
- Single Measurement Command
- Continuous Measurement Command
- Stop Measurement Command

06

Operating Instructions

Power-On Procedure

- Before powering on, connect the Laser Rangefinder Module, communication cable, DC power supply, and host controller according to the electrical interface requirements and as shown in the figure below.



- After the power supply is connected, pull the Power-EN pin high to activate the module.

Power-Off Procedure

- Before powering off, ensure that all operating processes and tasks have been completed and that the program has been exited properly.
- After confirmation, disconnect the power supply to power off the module normally.

Ranging Mode Operation

- Send the "Single Ranging" command to the Laser Rangefinder Module. The module performs a single ranging operation and returns the ranging status and distance value.
- Send the "Continuous Ranging" command to the Laser Rangefinder Module. The module performs continuous ranging and returns the ranging status and distance value continuously.
- Send the "Stop Ranging" command to terminate the ranging operation.

07

Precautions for Use

- Do not stare directly into the laser emission aperture to avoid eye injury (compliant with Class 1 standards).
- Operating temperature range: -20 ° C to +60 ° C. Avoid exposure to high temperature, humidity, rain, snow, or dusty environments.
- Do not disassemble the module without authorization, as this may void the warranty or cause device damage.
- This product is mainly intended for integration into civilian UAVs, outdoor observation, engineering surveying, and related equipment. This product is primarily designed for civilian and commercial applications.
- The optical axis calibration mode is intended for development personnel only. Do not aim the laser at human eyes while this mode is enabled to avoid eye injury.

08

Optical Window Requirements

- **Material Recommendation**

It is recommended to use **H-K9L** optical window manufactured by Chengdu Guangming Optics or fused silica as the optical window material. **H-K9L** and fused silica are the most commonly used optical-grade transparent materials, suitable for laser applications within the 300 nm to 2100 nm wavelength range, offering an excellent cost-to-performance ratio and excellent physical properties.

Machining Recommendation

The wedge angle tolerance of the optical window should be minimized as much as possible. A wedge angle tolerance of ≤3′ (tolerance grade ≤7) is recommended.

The optical surface should be as smooth as possible. A surface roughness with an arithmetic average deviation (Ra) of 0.012 is recommended.

Coating Recommendation

For 905 nm Laser Rangefinder Modules, it is recommended that the optical window be coated with an anti-reflection (AR) coating optimized for the 855 nm−955 nm wavelength range, with transmittance ≥99.5%.

Depending on the specific operating environment, additional protective coatings such as hydrophobic coatings or hard coatings may be applied to the outer surface of the optical window. Other optical specifications should comply with applicable industrial optical standards. requirements, with overall transmittance ≥98%.

Optical Window Structure and Installation Recommendation

The effective aperture of the optical window depends on the specific product model. The external dimensions of the optical window shall satisfy the following requirements:

Effective aperture of optical window − outer diameter of optical window ≥2 mm

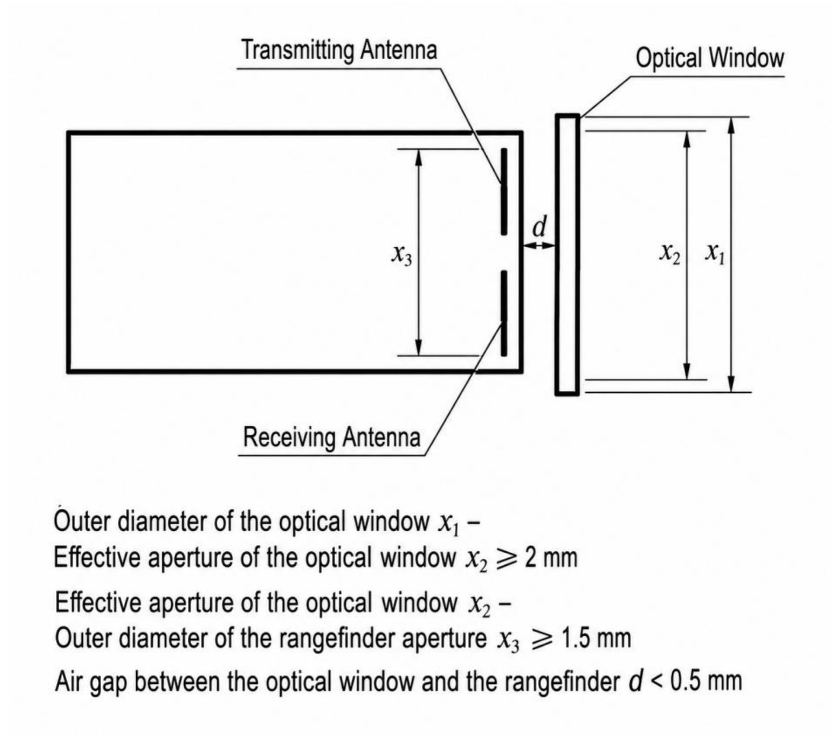
Outer diameter of rangefinder aperture − projected size of optical window effective aperture ≥1.5 mm

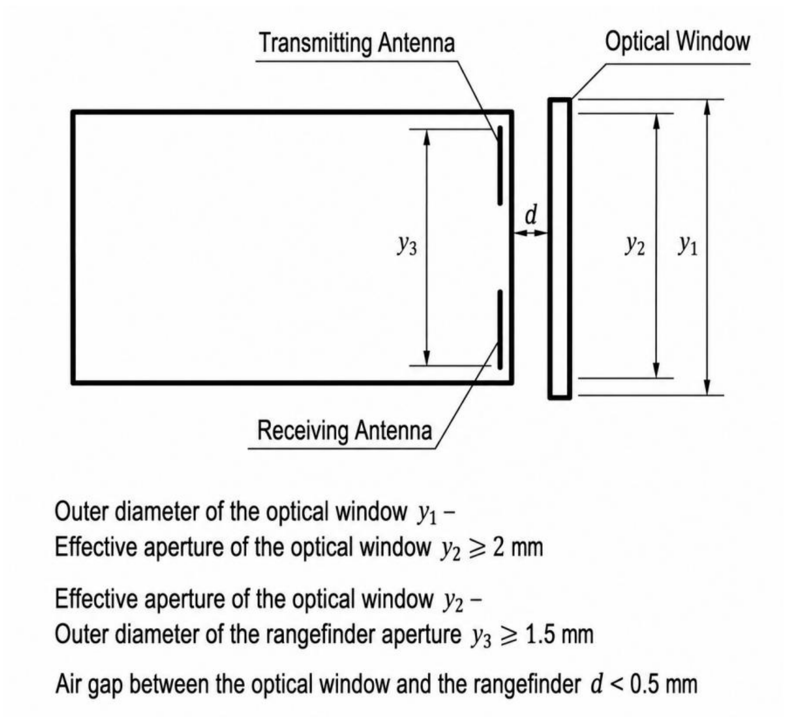
A schematic diagram is shown below.

Since the optical window absorbs a certain amount of laser energy, it is recommended that the window thickness be controlled within 2–4 mm according to the overall dimensions.

Due to the high transmittance of the optical window, it is recommended that the transmitting optical axis remain parallel to the normal axis of the optical window. A schematic diagram of the positional relationship between the optical window and the dual optical barrels is shown below.

At the same time, the air gap between the optical window and the Laser Rangefinder Module should be less than 0.5 mm. The figure below illustrates two recommended optical window installation methods.





09 Safety Precautions

- The module complies with IEC 60825-1 Class 1 laser safety requirements under normal operating conditions.
- Although classified as Class 1, prolonged direct staring into the laser emission aperture is not recommended.
- Safety design and analysis are conducted in accordance with GJB 900A-2012, General Requirements for Equipment Safety;
- Non-flammable materials are used, and all mechanical and electrical interfaces are securely connected;
- Appropriate design measures are implemented to prevent moisture accumulation that could result in short circuits;
- The module operates below the safety voltage limit for the human body.

10 Maintenance Instructions

Packing List

The standard items supplied with the product are listed in the table below. Optional accessories may not be included in this list and shall be subject to the actual product delivered.

No.	Item	Quantity	Remarks
1	Laser Rangefinder Module	1 unit	-
2	Certificate of Conformity	1 copy	-
3	Packaging Box	1 pc	-
4	Power Cable	1 pc	-

Module Cleaning

Cleaning of Optical Components

- Dust particles should be removed using an air blower;
- Fingerprints should first be cleaned with absorbent cotton lightly moistened with a small amount of alcohol-ether mixture, and then wiped with a clean lens cloth.

Cleaning of Mechanical and Electronic Components

- With the power disconnected, gently wipe mechanical and electronic components with alcohol and allow them to dry naturally before use;
- Keep the module, connectors, and cables away from moisture and contaminants whenever possible;
- Ensure that the equipment is completely dry before packaging.

Inspection and Maintenance

Routine Maintenance

When the product is used for the first time, or after replacement of related modules or components, both visual inspection and power-on inspection shall be performed. For products under normal operation, only a power-on inspection is required before use.

- Visual Inspection Procedure

- Check whether the product appearance is normal;
- Check whether the cable connections are correct and securely connected.

- Power-On Inspection Procedure

- Perform the power-on procedure and verify that the standby current is correct;
- Start the self-test function of the product;
- After the inspection is completed, perform the power-off procedure.

Periodic Maintenance

Under normal operating conditions, the Laser Rangefinder Module does not require routine maintenance. If the module has been stored in a dust-free environment for more than one year, maintenance should be performed, including general inspection and power-on inspection.

- General Inspection (Performed with Power Off)

- All markings and serial numbers on the product and test cable connectors shall be correct and clearly legible;
- All screws on the panel shall be securely fastened;
- The optical window of the product shall be free from visible defects that may affect normal operation, such as spots, pitting, water stains, fungus, fingerprints, dust particles, or cracks.

- Comprehensive Power-On Inspection and Maintenance

- Connect the product power supply in sequence;
- Perform the power-on procedure and verify that the standby current is correct;
- Start the product self-test and complete the self-test procedure;
- Perform the power-off procedure after completion of the inspection.

Troubleshooting

The following list only includes possible fault phenomena not caused by the product itself. If any other technical problems occur during operation, please contact our company for technical support.

Problem / Symptom	Possible Cause	Troubleshooting Method	Diagnostic Criteria	Solution
No communication or abnormal communication with the module	Power supply issue	Check whether the power supply has hardware or software wake-up enabled, verify voltage settings, and ensure the current protection setting is correct. If the voltage is correct, try replacing the power supply.	If the module operates normally after changing the voltage or power supply, the fault is determined to be caused by the power supply.	Set the correct voltage/current or replace the power supply.
	Cable connection issue	Check whether the wiring is secure or replace the communication cable (TTL communication cables should not be excessively long).	If communication is restored after reconnecting or replacing the cable, the fault is determined to be caused by the cable connection.	Reconnect or replace the communication cable.
		Check whether the wiring matches the interface definition.	If the module operates normally after correcting the wiring, the fault is determined to be caused by incorrect wiring.	Rewire according to the correct interface definition.
	Incorrect baud rate setting	1. Check whether the module status and baud rate settings match the communication protocol. 2. If baud rate configuration is supported, set the required baud rate and test communication again.	If communication becomes normal after correcting the baud rate, the fault is determined to be caused by an incorrect baud rate setting.	Set the host controller baud rate according to the communication protocol.
	Incorrect communication command	Check whether the sent commands match the communication protocol and confirm that commands are transmitted in hexadecimal format.	If communication becomes normal after sending correct commands, the fault is determined to be caused by incorrect communication commands.	Send the correct commands to the module.
	Main control program issue	If communication fails during integration with the main controller, verify whether standalone communication is normal.	If standalone communication is normal, the fault may be caused by the main control program.	Check the system software, including checksum settings and host controller data parsing/display functions.

	Other causes (circuit fault, etc.)	Return to the manufacturer for inspection.	To be determined by manufacturer inspection.	Contact the manufacturer for technical support or repair service..
Invalid ranging data returned	Operational issue	Check whether the transmitting or receiving lens is blocked, and ensure that there are no obstructions in the ranging environment (such as window glass or wall corners).	If ranging returns to normal after removing the obstruction, the fault is determined to be caused by blockage.	Remove the obstruction.
		Check whether the ranging axis and sighting axis are aligned.	If ranging returns to normal after realignment, the fault is determined to be caused by axis misalignment.	Realign the optical axes.
	Laser failure	Check whether the operating current is normal, observe whether the current fluctuates during ranging, send a self-test command, or use an infrared test card/CCD camera to observe whether a laser spot is present.	If the self-test indicates abnormal laser output and no spot is observed, the fault is preliminarily determined to be caused by laser failure.	Contact the manufacturer for technical support or repair service..
	Other causes (circuit fault, detector issue, etc.)	Return to the manufacturer for inspection.	To be determined by manufacturer inspection.	Contact the manufacturer for technical support or repair service..
False alarm during ranging	Electrical noise interference	Block the receiving lens or point the module toward the sky (away from direct sunlight) and check whether false alarms persist. Test the module with separate power supply and communication configurations to identify noise sources.	If false alarms disappear under isolated test conditions, the fault source can be identified as optical, communication, or power noise interference.	Reduce noise interference, improve grounding/filtering, use ferrite cores, or increase threshold settings (may reduce ranging capability).
Insufficient ranging capability	Weather conditions	Check whether weather conditions such as rain, snow, fog, dust, low visibility, high humidity, or	If ranging capability meets specifications under proper weather conditions, the fault is	Test under suitable weather conditions.

		atmospheric turbulence are present.	determined to be caused by environmental conditions.	
	Optical window issue	Check whether the optical window has low transmittance, incorrect wavelength coating, contamination, small aperture, or excessive tilt angle. Remove the optical window and retest.	If ranging capability returns to normal after removing the optical window, the fault is determined to be caused by the optical window.	Replace the optical window according to the recommended specifications and installation requirements.
	Target size too small or reflectivity too low	Replace the target with a larger or higher-reflectivity object, preferably perpendicular to the optical axis.	If ranging capability improves with a more suitable target, the fault is determined to be caused by improper target selection.	Use a more suitable target or select a higher-performance module.
	Operating temperature out of range	Send a self-test command and check the module operating temperature. Retest after returning to normal temperature.	If ranging capability returns to normal after temperature recovery, the fault is determined to be caused by excessive operating temperature.	Operate within the specified temperature range.
	Optical axis deviation	Confirm whether the module has undergone vibration, shock, temperature cycling, or drop events.	To be determined by manufacturer inspection.	Contact the manufacturer for technical support or repair service..
	Other causes (circuit fault, low receiver sensitivity, etc.)	Return to the manufacturer for inspection.	To be determined by manufacturer inspection.	Contact the manufacturer for technical support or repair service..
Accuracy out of tolerance	Large target reflectivity deviation	Confirm whether the test target has excessively high reflectivity and whether the returned data fluctuates significantly.	If accuracy returns to normal after changing the target, the fault is determined to be caused by target reflectivity.	Replace the target. High-reflectivity targets are not recommended.
	Improper test operation	Check whether the target is correctly aligned.	If accuracy returns to normal after proper alignment, the fault is determined to be caused by improper operation.	Realign the target.

	Calibration instrument issue	Verify the accuracy of the calibration instrument.	If accuracy returns to normal after replacing the calibration instrument, the fault is determined to be caused by inaccurate calibration equipment.	Replace the calibration instrument.
	Low optical window transmittance	Check whether the returned ranging value is excessively large. Remove the optical window and retest.	If accuracy returns to normal after removing the optical window, the fault is determined to be caused by low optical window transmittance.	Replace the optical window according to the recommended specifications and installation requirements.

Packaging and Transportation Requirements

Packaging

After unpacking, if the product needs to be placed back into storage, it shall be repackaged using the original packaging whenever possible.

If the product needs to be returned to the manufacturer, the original packaging is recommended. If alternative packaging is used, it shall not cause any degradation in product performance or physical damage.

Transportation

Repackaged products may be transported by automobile, railway, aircraft, or ship. During transportation, the package shall be securely fixed to the transport vehicle to avoid impact, improper handling, and exposure to rain or snow.

11

Technical Support and Service

SunFlaser Technology provides a one-year limited warranty and lifetime technical support from the date of delivery.

During the warranty period, defects caused by manufacturing or material issues will be repaired or replaced free of charge.

Damage caused by improper operation, unauthorized modification, misuse, or external factors is not covered under warranty.

12 Contact Information

Website: www.sunflaser.com

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

Appendix 1：

1 Communication Protocol

1.1 Communication Interface

Baud Rate: 115200 bps (default), 9600, 14400, 19200, 38400, 57600, 115200, 230400 bps

Single-Byte Transmission Format:

1 start bit, 8 data bits, no parity bit, and 1 stop bit.

For 8-bit data transmission, UART data is transmitted LSB first..

Distance Data Format

Distance = (DATA_H << 8) | DATA_L

Unit: meter (m)

1.2 Single Measurement Command

Note:

Checksum Calculation Method

Checksum = (Byte3 + Byte4 + Byte5 + Byte6 + Byte7) & 0xFF

Receive checksum = (Byte1 + Byte2 + Byte3 + Byte4 + Byte5 + Byte6 + Byte7) & 0xFF

Command Sent to the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x88	0xFF	0xFF	0xFF	0xFF	Checksum

Response Returned by the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x88	Status	0xFF	DATA_H	DATA_L	Checksum

Status = 0: Single measurement failed; DATA_H = 0xFF, DATA_L = 0xFF

Status = 1: Single measurement successful

DATA_H = High byte of measurement result

DATA_L = Low byte of measurement result

1.3 Continuous Measurement Command

Note:

Checksum Calculation Method

Checksum = (Byte3 + Byte4 + Byte5 + Byte6 + Byte7) & 0xFF

Receive checksum = (Byte1 + Byte2 + Byte3 + Byte4 + Byte5 + Byte6 + Byte7) & 0xFF

Command Sent to the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	Command ID	0xFF	0xFF	0xFF	0xFF	Checksum

Response Returned by the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	Command ID	Status	0xFF	DATA_H	DATA_L	Checksum

Status = 0: Continuous measurement failed; DATA_H = 0xFF, DATA_L = 0xFF

Status = 1: Continuous measurement successful

DATA_H = High byte of measurement result

DATA_L = Low byte of measurement result

Command ID / Mode Definitions

Command ID = 0x89: Continuous ranging mode

Command ID = 0xF9: Optical axis calibration mode

(The module returns the calibration status once after receiving the calibration command.)

1.4 Stop Measurement Command

Command Sent to the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x8E	0xFF	0xFF	0xFF	0xFF	Checksum

Response Returned by the Rangefinder Module

Byte	1	2	3	4	5	6	7	8
Description	0x55	0xAA	0x8E	Status	0xFF	0xFF	0xFF	Checksum

Status = 0: Failed to stop continuous measurement

Status = 1: Continuous measurement stopped successfully